1. This is an individual project. Derivations and coding should be the sole work of the individual submitting the work. Discussions with TA, faculty and peers in the class are permitted. Use of textbook is also permitted. Use of generative AI to debug the code is allowed. The student should clearly note how this resource was used in the submission. However, no online tutors (for example, tutors on sites like chegg.com) or other sources for any other part such as formulation of the project may be used.
2. You are required to use Matlab or a programming language to solve the problem. Use of *Excel, or a commercially available solver is not permitted*.
3. Begin your project as soon as it is assigned. Do not wait until the last minute. Formulation and debugging of the code can take longer than you may expect.
4. Submit a single pdf of derivation/formulation and the code and solution outputs in gradescope. In a separate gradescope assignment “Project 1 code” submit your code.

**Problem statement**

Luke Skywalker is practicing with his lightsaber. The movement of the lightsaber in air generates a convective heat transfer coefficient of h = 50 W/m$^2$-K. The freestream air is at $T_{air}$ = 20 ºC. The lightssaber consists of a laser core that is $L_1$=120 cm long and of [illegible] $R_{core}$ = 2 cm. The thermal conductivity of the core is $k_{core}$ = 0.35 W/m-K. The core generates volumetric heat non-uniformly as given by the expression below

$R_{core}$ is the radius

$$\dot{q}'''(r,x) = \dot{q}'''(r=0)\cdot\left(1-\left(\frac{r}{R_{core}}\right)^2\right)$$

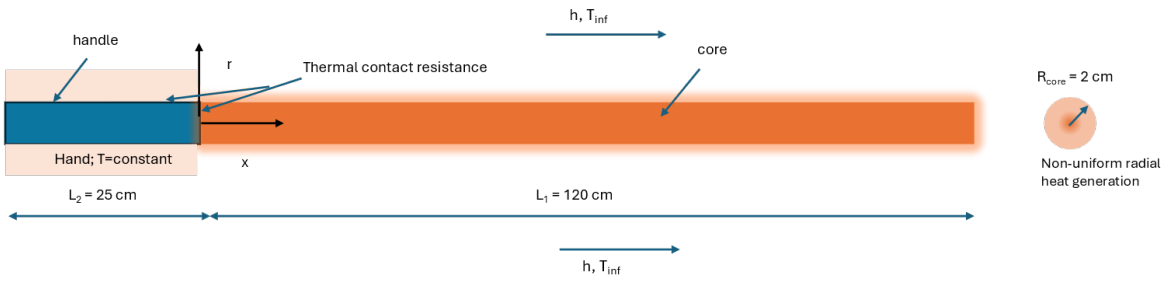
where $\dot{q}'''(r=0) = 10^6\ W/m^3$ and x=0 is at the intersection of the laser core and the insulating handle.

The handle of length $L_2$ = 25 cm is made of a low thermal conductivity ceramic ($k_{handle}$ = 0.05 W/m-K). There is a contact resistance between the handle and the laser core of $R''_{tc} = 10^{-5}$ m$^2$-K/W. The figure shows the details including dimensions and boundary conditions.

Assume the entire handle of the sword is gripped firmly by Luke Skywalker and his hands are at a constant of 37.1 ºC and there is a thermal contact resistance of $R''_{tc} = 10^{-4}$ m$^2$-K/W between the sword's handle and Luke's hand.



Credit: Star Wars, Luke Skywalker - Free Wallpaper / WallpaperJam.com



(1) Develop a 2D steady state numerical conduction model for the lightsaber.

Identify the unique control volumes for which different energy balance equations need to be developed. Derive these equations for unique control volumes.

You need to have at least 5 nodes in the radial direction for the core and insulation regions. In the axial direction, use a minimum of 12 nodes in the axial direction for the core, and a minimum of 5 nodes in the axial direction for the handle.

(2) Solve the system of equations to determine the temperature distribution within the lightsaber.

(3) Plot a contour of the temperature distribution in the core and handle.

(4) Determine the rate at which heat transfer is transferred to Luke's hand from the laser core1.

(5) Out of frustration, Luke plunges the lightsaber into a lake whose temperature is $T_{lake}$ = 10 °C. The intense heat from the lightsaber causes the water surrounding the lightsaber to boil. Determine the new temperature distribution within the lightsaber. Assume a convective of heat transfer coefficient of 5,000 W/m$^2$-K during the boiling process.

**Grading Rubric**

| Section | Points Allocated |
|---|---|
| 1. Sketch identifying the origin, coordinate system, nodal numbering scheme, and unique control volumes for which energy balance equations need to be derived | 1 |
| 2. Derivation of nodal equations for unique control volumes and pseudo code | 8 |
| 3. Code completeness and accuracy | 4 |
| 4. Contour plot of temperature (#3) with brief discussion of plot | 1 |
| 5. Parts #4 and #5 | 1 |

More assumption:

=> Luke's hand covers the base of the handle as well.

=> symmetry across r=0

=> left side handle is in contact with hand